JOOD A. AL ASWAD

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PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow, Virginia Tech, Stratigraphic Paleobiology Lab 2025–present

EDUCATION

Ph.D. Geological Sciences: Quantitative Paleobiology, Stanford University 2025

M.S. Geological Sciences: Stratigraphy & Petrophysics, Cornell University 2019

B.S. Earth Science, George Mason University 2016

GRANTS AND FELLOWSHIPS

Thomas D. and Janice H. Barrow Graduate Fellowship (~$56,000) 2024–2025

Graduate Fellowship, King Abdullah Scholarship Program: Stanford (~$205,000) 2019–2022

Graduate Fellowship, King Abdullah Scholarship Program: Cornell (~$150,000) 2017–2019

Donovan Family Scholarship: Cornell ($2,000) 2018

Full Ride Scholarship, King Abdullah Scholarship Program 2011–2016

Undergraduate Student Research Grant, URSP, George Mason University (~$1,500) 2015

SCIENTIFIC PUBLICATIONS

8. **Al Aswad, J.,** Bazzi, M., Penn, J., Monarrez, P.M., Deutsch, C., and Payne, J. Coupling of climate change and extinction influences taxonomic homogenization in deep time. *Proceedings of the Royal Society B.* *In press.*

7. **Al Aswad, J.,** Ashing-Giwa, K. and Monarrez, P.M., Thermal tolerance determines marine bivalve invasion success in California across 120,000 years and points to the next arrivals. *In prep.*

6. **Al Aswad, J.,** Hautmann, M., Singh, P., Ferrill, N.L., Al-Ramadan, K., Lehrmann, D.J., Morsilli, M.D., Koeshidayatullah, A.I., and Payne, J.L. Bivalves from the Lower Triassic of central Saudi Arabia: Systematics and calibration of Permo-Triassic boundary position within the Khartam Member of the Khuff Formation. *In prep.*

5. Bazzi, M., Lloyd, W.L., Ebersole, J., Sternes, P., **Al Aswad, J.,** and Payne, J. Extinction threatens to cause morphological and ecological homogenization in sharks. *Science Advances. DOI: 10.1126/sciadv.aea0278.* 2025

4. **Al Aswad, J.,** Penn, J., Bazzi, M., Monarrez, P.M., Deutsch, C., and Payne, J. Physiology and climate change explain unusually high similarity across marine communities after end-Permian mass extinction. *Science Advances. DOI: 10.1126/sciadv.adr4199* 2025

3. Payne, J.L., **Al Aswad, J.,** Deutsch, C., Monarrez, P.M., Penn, J.L., and Singh, P. Selectivity of mass extinctions: Patterns, processes, and future directions. *Cambridge Prisms: Extinction. DOI: 10.1017/ext.2023.10* 2023

2. Tester, J., Beyers, S., Gustafson, J.O., Jordan, T.E., Smith, J.D., **Al Aswad, J.A.,** et al. District geothermal heating using EGS technology to meet carbon neutrality goals:

A case study of Earth Source Heat for the Cornell University campus. *Proceedings World Geothermal Congress.* 2020

1. Gustafson, J.O., Smith, J.D., Beyers, S.M., **Al Aswad, J.A.,** Jordan, T.E., and Tester, J.W. Earth Source Heat: Feasibility of deep direct use of geothermal energy on the Cornell campus. *GRC Transactions, Vol. 4*.

2018

HONORS AND AWARDS

Research and Innovation Postdoctoral Fellowship, Virginia Tech 2026 – present

Winifred Goldring Award (Outstanding Paleontology PhD Student), 2025
Association of Woman Geoscientists and the Paleontological Society

Centennial Teaching Assistant Award, Stanford University 2025

Harriet Benson Fellowship Award for Excellent Research, Stanford University 2022

Best Scientific Video: Storytelling, Stanford BioX 2020

Certificate for Highest GPA in Program, George Mason University 2016

Outstanding Senior in Earth Science Award, George Mason University 2016

TEACHING EXPERIENCE

Guest Lecturer: GEOS 5984 Advanced Geosciences Data Analysis with R, Virginia Tech 2026

Teaching Assistant: EPS 169/269 The Sixth Mass Extinction, Stanford University 2024
Spring 2023
Autumn 2023

Teaching Assistant: EPS 136/236 Macroevolution, Stanford University 2024

Guest Lecturer: Physical Geography (Climate Change), Foothill College 2023

Teaching Assistant: GEOL 4 Coevolution of Life and Earth, Stanford University 2022

Module Instructor: Introduction to Paleobiology, Friday Harbor Laboratories 2021

WORKSHOPS AND EDUCATIONAL TRAINING

Introduction to Earth System Modeling Workshop 2026
UC Riverside · Instructor: Dr. Andy Ridgwell

Best Practices in Teaching and Mentoring Workshop 2026
Virginia Tech Center for Teaching and Learning

Stanford Course Design Institute 2025

Invertebrate Zoology Course 2023
Grade: 99/100 · Washington University and Friday Harbor Labs

Geological Field Camp 2015
Grade: A · South Dakota School of Mines and Technology

STUDENT MENTORING

Pokémon-Inspired Evolution Museum Event 2026–present
Virginia Tech Museum of Geosciences. Co-created and mentored 4 graduate students in creating interactive public event teaching children about evolution and biodiversity.

Mentoring graduate students, Virginia Tech 2025–present

Undergraduate & high school research mentor and mentor assistant 2022
Stanford SURGE, Bio-X, and Stanford Earth Young Investigators programs: Victor Trujillo, Lucy Helms, Edward Huang, Kelly Tung, McKenna Sanders, Sakeena Saber.

Peer Advisor, Bay Area Graduate Pathways in STEM 2019–2020

LEADERSHIP AND SERVICE

Organizer, International Fossil Coral and Reef Society Early Career Symposium 2026- present
Creator, Oceanic Diversity and Rarity Exhibit at Virginia Tech Museum Event 2026- present
Creator & Committee Leader, DEEP Awards 2022–2023

Access, Belonging & Community Liaison, Stanford Doerr School of Sustainability 2024

Committee Member, The Paleontological Society 2021–2025

Student Representative, Paleontological Society Council 2021–2023

Organizer, Bay Area Paleontological Conference (Cancelled due to Covid) 2020

President, Cornell Earth Resources Energy Club 2018

Volunteer for 100+ Hours, George Mason University APO 2013-2016

OUTREACH FUNDING

Al Aswad, J. Funds for Grants Program, The Paleontological Society (~$10,000). 2023

Al Aswad, J. Grant for DEEP Paleontology Awards, The Paleontological Society (~$12,000). 2022

PRESENTATIONS

INVITED PRESENTATIONS

"The Great Dulling: climate, thermal physiology, and mass extinctions explain when and why global communities homogenize in deep time." Geological Society of America, Colorado. 2026

"Synergy of climate change and biodiversity loss drive global taxonomic homogenization in deep time." Departmental Seminar Series, George Mason University. 2026

"Examining the compositional similarity of marine invertebrate ecosystems after the Big Five mass extinction events." Fossil Coffee Series, UC Berkeley. 2024

"Taxonomic homogenization of marine ecosystems after the end-Permian mass extinction." Oxford University Biogeography Workshop, Oxford, UK. 2024

"Physiology of marine invertebrates explains broadened biogeography after end-Permian extinction." University of Lausanne, Switzerland. 2023

SELECTED CONFERENCE PRESENTATIONS. ** denotes student mentee*

Trujillo, V.*, Helms, L.*, Al Aswad, J.A., Monarrez, P., and Payne, J.L. "Biogeographic stability of benthic marine mollusks across the end-Cretaceous mass extinction." GSA Annual Meeting, Denver. 2022

Tung, K.*, Anderson, M.*, Saber, S.*, Monarrez, P.M., Al Aswad, J.A., and Payne, J.L. "Comparing the latitudinal ranges of genera of Mollusca and Arthropoda before and after the end-Permian mass extinction." AGU Fall Meeting, Washington D.C. 2022

Al Aswad, J.A. "Porosity, permeability, and paleotopographic analysis: A geological interpretation of geothermal reservoir quality at Cornell University." GSA Annual Meeting, Indianapolis. 2018

PEER REVIEW

Reviewer for: *Science; Science Advances; Gondwana Research; Palaeogeography, Palaeoclimatology, Palaeoecology; Cambridge Prisms: Extinction*

TECHNICAL SKILLS & COMPETENCIES

Statistical modeling & data science	R/RStudio (intermediate), MATLAB and Python (beginner)
Earth system modeling	cGENIE (climate dynamics, biogeochemical cycles)
Fossil identification	Marine invertebrate paleobiology (bivalves)
Field skills	Paleontological and stratigraphic field work, geological mapping
Design	Adobe Creative Suite, digital art
Languages	English (fluent), Arabic (conversational)